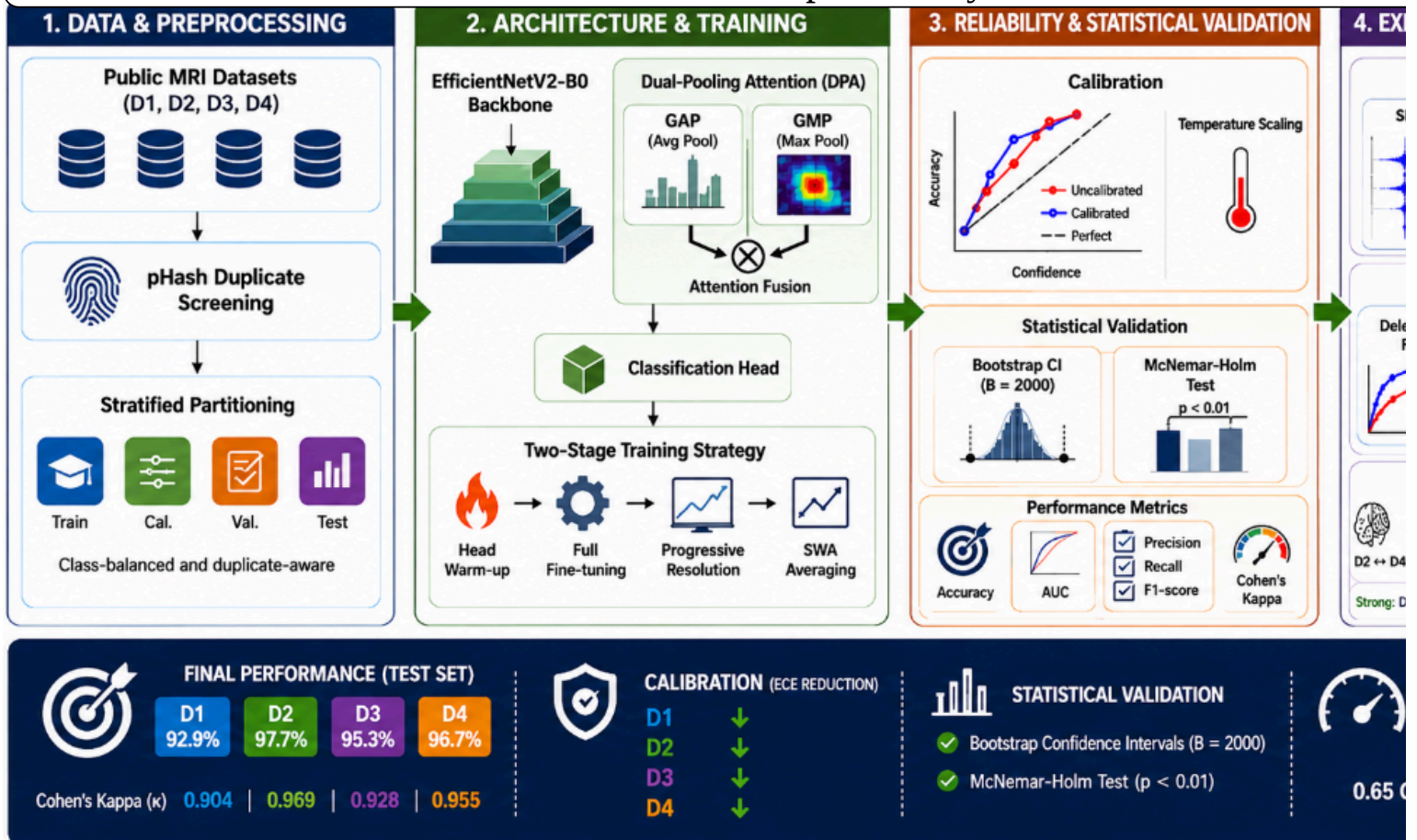


Beyond Accuracy: A Reliability-Oriented Framework for Brain Tumor MRI Classification with Calibration, Explainability, and Cross-Dataset Validation



Conclusion: The results demonstrate that calibration, statistical robustness, explainability, and generalization are complementary perspectives that are essential for the trustworthy assessment of medical image classification models.

MRI
tion

PLAINABILITY & GENERALIZATION

Tri-Method XAI Ensemble

iHAP

Grad-CAM++

Integrated Gradients

Explanation Validation

etion-Insertion Faithfulness

IoU Agreement (Cross-Method)

Cross-Dataset Transfer

MODEL EFFICIENCY

6.65M Parameters

GFLOPs | <10 ms / image

alization provide
ification systems.